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MSU-TSCH: A Mobile scheduling updated algorithm for TSCH in the internet of things

Wassim Jerbi, Omar Cheickhrouhou, Abderrahmen Guermazi, Hafedh Trabelsi

***Abstract*— On top of the low-power IEEE 802.15.4 radio, IEEE created IEEE 802.15.4e time slotted channel hopping (TSCH), an exceptionally efficient, reliable, and predictable time-frequency enabled medium access control (MAC) protocol for the industrial sector. However, the IEEE 802.15.4e TSCH specification does not address how scheduling would be created, updated, or maintained. Moreover, TSCH does not support mobility. Although several scheduling techniques have been proposed in the literature to fill this gap, these techniques did not consider networking metrics such as duty cycle, network capacity, end-to-end delay, and packet delivery ratio.**

In this paper, we develop a new dynamic scheduling approach for TSCH that supports mobility and adapts to topology changes, called mobile scheduling updated TSCH (MSU-TSCH). attempts to schedule time slots by first picking the closest nodes to the sink. We undertake comprehensive simulations to assess the effectiveness of MSU-TSCH with PRCOS, an existing scheduling algorithm in a mobile context. This proposal reduces the received energy consumption by up to 40%, on the other hand, it decreases the average end-to-end latency by 30%.

Index Terms— IEEE 802.15.4e TSCH, Internet of Things (IoT), Scheduling algorithms, MSU-TSCH, Industrial Wireless Sensor Networks (IWSN).

I. Introduction

A significant amount of new additional traffic is introduced by IoT to connect several devices to the internet and to make the environment more "intelligent". IoT applications include healthcare, smart cities, transport, and industry 4.0. Unlike Ethernet networks, radio connectivity in the IoT domain is unstable and does not meet industrial IoT applications requirements of low-power and stable connectivity.

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Several standard protocols have emerged in recent years to support IWSNs in various application domains, we mention for example WirelessHART [1], ISA100.11a [2], and IEEE 802.15.4e [3]. These protocols have also been tested in various industrial automation situations.

The IEEE 802.15.4 standard cannot meet the requirements of industrial applications. The IEEE 802.15.4e amendment was created as a result, and it offers innovative MAC-layer profiles that are tailored for a variety of application areas, such as process automation. The IEEE 802.15.4e standard offers three MAC modes which are the Deterministic and Synchronous Multi-channel Extension (DSME), The Low Latency Deterministic Network (LLDN) and the Time Slotted Channel Hopping (TSCH). The goal of TSCH is to schedule the communication [4].

TSCH allows nodes to communicate according to a schedule that determines access to each cell, i.e who can receive and transmit on a particular channel at a given time. Centralized schedule creation is possible, as demonstrated by the Network Manager in WirelessHART. This method allows increased operator control and network-wide resource reservations. Our focus is on schedulers designed with the IIoT in mind and suggested for the 802.15.4 TSCH MAC. Important networking parameters like duty cycle, network capacity, end-to-end delay, packet delivery ratio, etc., are fundamentally impacted by the resources scheduling. Therefore, the optimization of these metrics is essential to meet the demands put forth by industrial networks.

The following are the primary contributions of our study:

- ✓ Reduce the number of collisions that occur in shared slots, as well as packet loss and overall control traffic.
- ✓ Reduce the slotframe occupancy and length,
- ✓ Reduce the end-to-end delay: Calculate the average frames end-to-end delay passing through the network during a given period from the delay values for each frame collected, or for the frames collected of each source.
- ✓ Reduce energy consumption: channel access is expensive in wireless networks, and energy consumption increases with the amount of information to be transmitted. Therefore, the number of control messages and their size should be reduced.

We present MSU-TSCH, a centralized communication scheduling technique for TSCH networks. In a mobile and static context, we compare MSU-TSCH with PRCOS [5] communication scheduling algorithm.

The reminder of this paper is as following. The related work is

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included in Section 2. Sect. 3 contains the network model. The MSU-TSCH algorithm that we propose is described in Section 4. Sect. 5 contains the MSU-TSCH performance evaluation. Sect. 6 offers the conclusion as well as plans for future work.

II. RELATED WORK

TSCH is frequently used in wireless networks to enable them to comply with the stringent industrial domain requirements. TSCH is based on TDMA and FDMA as its foundations and is suitable for obtaining access determinism and low-power communication in IWSNs. The original purpose of this protocol was to improve QoS and to avoid collisions through the congested 2.4 GHz wireless band by providing time division and frequency division multiple access. Additionally, TSCH is reliable in terms of energy consumption because the communication devices of the sensor are only activated when they are asked to communicate, and the timing is precisely programmed. However, TSCH scheduling is not defined and is application dependent. Indeed, the standard just offers a framework and makes no requirement on the scheduling mechanism.

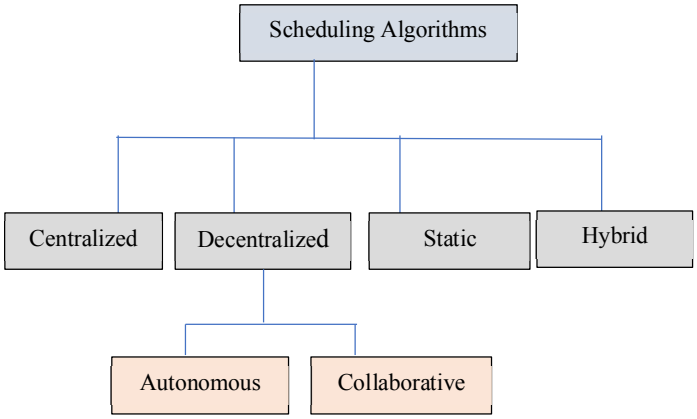

Fig. 1 Classification of TSCH Scheduling Algorithms

Scheduling algorithms should adapt to frequent changes in network features, such as nodes mobility, radio environments shifting, or updated application needs, without incurring a considerable expense.

Consequently, several researches addressed the TSCH scheduling problem and proposed new algorithms.

As shown in figure 1, the existing scheduling algorithms can be classified into four categories according to their operation: Centralized, Decentralized, Static and Hybrid.

Decentralized (called also distributed) scheduler can be autonomous or collaborative. Among decentralized autonomous scheduling algorithms, we cite DIGEST [6], which adapt to variable traffic (refers to the scheduler's ability to adapt to traffic volume changes). Decentralized collaborative approaches depend on nodes information exchange. However, few collaborative schedulers consider the efficiency metric. These exchanges are susceptible to delays and breakdowns, much like regular traffic, which could have an effect on the effectiveness of a scheduler [7].

Static schedulers were proposed for a smart metering application where a large number of devices periodically send data. The suggested static scheduler is comparable to a slotted ALOHA [8] method, where reliability and latency goals are met at the expense of increased energy usage.

To avoid previous schedulers flaws, hybrid approaches were proposed. Hybrid scheduling is a merge of collaborative, static, and autonomous approaches. For example, PAAS [9] and e-TSCH-Orch [10] autonomous schedulers did not support heterogeneous traffic. To combat this, they add the Orchestra cooperative method, enabling nodes to exchange scheduling modifications as traffic intensity changes.

When traffic, topology, and the environment are less variable and rescheduling is rarely required, centralized schedulers may be favored. This is what we will show through our analysis of centralized scheduling and the advantage it provides in terms of reliability. With a global knowledge of the network, centralized TSCH scheduling algorithms try to create an ideal timetable.

Centralized scheduling algorithms benefit from fully knowledge of the state of the network. While some algorithms rely on the gathering of statistics during network startup, others implicitly assume that this knowledge is acquired before the network is deployed [11]. The scheduling algorithm in this instance resembles an optimization problem as it attempts to minimize (or alternatively maximize) a specified metric. The first pioneer work in TSCH scheduling is Traffic-Aware Scheduling Algorithm (TASA) [12], which is a centralized convergecast solution that employs matching and coloring processes. The communication schedule in TASA is created by a central gateway based on the topology of the tree network and the amount of traffic. When the routing topology changes dynamically, the central element must rebuild the timetable from scratch. As a result, centralized systems like this are unsuitable for dynamic networks.

Adaptive MUlti-hop Scheduling method (AMUS) [13] assigns Transmission slots based on the assumption that nodes with fewer hops to the root have more traffic to forward. First, AMUS creates a three-dimensional matrix with axes for frequency offsets, node IDs, and timeslots. Each component has a TSCH cell assigned to it.

Second, each route matching cells are placed in the matrix in the order specified in the Scheduling Sequence (SS). The method for setting SS, a manually set initial matrix, was not suggested in the original paper. In other words, a node closer to a sink node communicates more since AMUS secures mutually connected cells for every single route. AMUS does not the hotspot unsolved problem. Forwarding and TSCH Scheduling (FTS-SDN) [14] which handles node mobility and transmission scheduling in IWSN providing bounded end-to-end delays.

PRuning-based COloring Scheduling (PRCOS), based to about applications for data collecting. This algorithm optimizes the lifespan of the node that consumes the most power, in order to increase the lifetime of the WSN.

III. TSCH SCHEDULE

A. Transmission schedule TSCH

We consider a multi-hop industrial network with relays, deployed over a large area using the IEEE 802.15.4-2015 standard and the 6TiSCH stack [15], with a mesh or tree topology. Typical values proposed in existing implementations of IEEE Std 802.15.4-2015 TSCH mode are, timeslots of 10 ms

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duration, maximum frame size of 127 bytes, throughput of 250 kbps and slotframe size of several hundred timeslots [16].
For the transmission to take place correctly in a fixed size interval, the application messages that are too long must be fragmented so that each fragment constitutes a frame of size smaller than the maximum, which will be transmitted on a cell.

Routing protocols become unstable as a result of the continuous topology changes caused by mobile nodes. The parent node switching process takes long time in traditional routing solutions like Routing Protocol for Low-Power and Lossy Networks (RPL)[17].

This study introduces Mobile Scheduling Updated TSCH (MSU-TSCH), which takes advantage of TSCH capabilities to increase mobile node support and get them connected to closer fixed nodes (or Static nodes). A parent node update may be done instantly and dynamically using MSU-TSCH.

B. Operation of TSCH scheduler

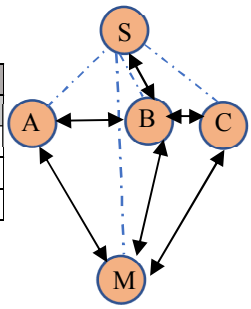
At each timeslot, a node may send a frame, receive a frame, or go into sleep mode to save energy. Depending on the quantity of produced packets and its position in the network, a node may participate in numerous timeslots per slotframe. The number of timeslots since the network was founded is identified with an Absolute Sequence Number (ASN) for each time slot.

Furthermore, in each timeslot, a single frame transmission is permitted, followed by a reception acknowledgement. TSCH schedulers are in charge of organizing transmissions and assigning one or more channel offsets per node and per link [18].

Figure 2 shows a topology composed of four fixed nodes (S, A, B, C) and a mobile node M. The mobile node M travel through a circular trajectory. In this example, only Node B communicates with the Sink.

The data packets transferred from the mobile node M to the Sink with various combinations of beacon periods and report periods are shown in Figure 2. In this TSCH scheduling sample, the number of timeslots by slotframe is thirteen. The communication is scheduled on 4 different channel offsets. The scheduling outcomes mostly depend on the number of hops a packet must travel to reach its destination. Node M has three paths to reach the sink S, which are: M-B-S, M-A-B-S or M-C-B-S. Indeed, for the M-B-S path, the transmission of inputs from nodes M and B must wait their slotframes, i.e, the transmission M-B occurs in the timeslot 12 and B-S in the timeslot 11. Therefore, the total transmission requires at least thirteen timeslots. Both paths M-C-B-S and M-A-B-S have the same number of hops. The total transmission of M-C-B-S requires at least six time slots (M-C-B = 3 timeslots, B-S = 3 timeslots), whereas M-A-B-S therefore requires at least fourteen time slots (M-A-B = 6 timeslots, B-S = 9 timeslots). Therefore, the M-C-B-S routing path present reduced delay than M-B-S. Therefore, we have to choose a better route between M and S to reduce delay.

Sink	
Routing	TSCH Schedule
M->B->S	(12,2) (11,3)
M->C->B->S	(6,4) (8,2) (11,3)
M->A->B->S	(9,4) (2,3) (11,3)



Channel offsets \ Slot frame	1	2	3	4	5	6	7	8	9	10	11	12	13
	A->all				S->B				B->C				S->all
				B->all				C->B				M->B	
		A->B	B->M			B->A	C->all				B->S		
		C->M			A->M	M->C			M->A	M->all			

Time slot

Fig. 2: Example of a FTS-SDN Scheduler

IV. Proposed MSU-TSCH

MSU-TSCH is proposed in this paper, which benefits from TSCH advantages. Using MSU-TSCH, a parent node change may be done instantaneously and dynamically. There are no limits to parental shifting or communication traffic. The MSU-TSCH technique extends TSCH to accommodate mobility while maintaining high reliability and real-time capabilities in terms of end-to-end delay.

MSU-TSCH protocol enables the transmitter to send packets on a certain channel at a given timeslot, and the receiver to receive these packets on the same channel. The link is a broadcast link if several receivers receive on the same channel in the same time slot. The MSU-TSCH provides each connection a fixed offset that allows the first entry in the list to be picked to reduce link collisions, which occur when two nodes broadcast on same channels in same timeslot.

MSU-TSCH configures the assignment between connections and devices when a new device joins the network. When a network event occurs, the sink describes the network's behavior in the context of decisions known as flow rules. These decisions are sent to nodes using so-called flow rules, which specifies how a node treat packets according to specific characteristics.

In order to maintain an up-to-date image of the network and respond promptly to changes, the Sink must receive regular communications, known as reports, that define, for every network node, the set of neighbour nodes and also the Received Signal Strength Indication (RSSI) of these neighbours. Each node sends its reports to the sink via a Virtualization Connection, which is a network made up of several nodes that act as repeaters as shown in figure 3.

Each node waits for beacon frames with the hop count towards the Sink, which is the node functioning in order to construct Virtual connections. The Sink initiates the process by broadcasting a beacon frame indicating its involvement in the network. The nodes near the Sink will begin broadcasting the beacon frame as soon as they receive it, indicating that they are directly linked to the Sink. To deliver its reports, each node uses a Virtual Link, which is a link made up of several nodes

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functioning as repeaters in the direction to the sink. These reports can be used by the sink to regulate TSCH network routing (By energy level of battery and RSSI). The network management of the MSU-TSCH approach leverage the sink network architecture to discover the best pathways.

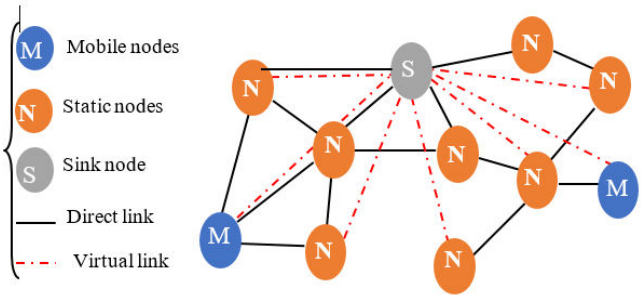

Fig. 3: The MSU-TSCH architecture

The management networks may also choose when and how to do parent handovers, as well as how and when to schedule all transmissions. The beacon and report periods can be set by the sink: the network manager. The management functions of sink are described in this study to allow mobility in the TSCH network. The MSU-TSCH determines which channels and timeslots the links that will be used for broadcast. There are three types of timeslots [14]:

- Uplink timeslots reserved for the unicast communication from nodes to the Sink.
- Downlink timeslots, reserved for the unicast communication from the Sink to a node.
- Broadcast timeslots, reserved for broadcast communications. When a node transmits a packet in its own broadcast timeslot, all the neighbor nodes will be listening on the same channel to receive the packet.

The schedule is written as a list of:

- ✓ transmission timeslots for the node.
- ✓ time intervals during which the node must listen for incoming transmissions.
- ✓ the channels that will be used for each of these tasks.

We consider three types of nodes: sink, fixed node, and mobile node. The Sink node's primary responsibility is to serve as the network manager. As a result, the Sink node should be connected to all other nodes. Fixed nodes, on the other hand, act as a relay node to the Sink and forward packets from and to the Sink. Mobile nodes move around fixed nodes and might act as a relay node for the fixed nodes.

The network management packets are always transmitted in the broadcast timeslots to prevent interfering with data packets. Each node route request is answered by the sink. When a node does not have any flow rules to classify a packet in a particular flow, it sends a route request to the sink.

Each node serves as a router, and therefore help to route packet. The nodes do this by checking each incoming or outgoing packet against the flow rules they receive, as determined by the sink. The nodes achieve this by comparing each incoming or outgoing packet against the management networks' flow rules.

The MSU-TSCH is able of dealing with common changes in the industrial environment for fixed nodes, such as allowing the network designer to build up redundant connections for transmitting packets over several pathways if the topology changes. The issue of mobile nodes is to interact with other nodes without making any assumption about their mobility. To achieve this goal, we add mobility support to TSCH by allocating one (channel, timeslot) pair for uplink communications and another pair for downlink communications with each stationary node.

Figure 4 shows the route establishment process. The Sink periodically broadcasts a beacon frame to establish/update routes. Upon reception, neighbors nodes to the Sink broadcasts this beacon frames by indicating their hop count to the sink. Parent nodes collect reports from its child nodes and aggregate them, then they forward to the sink. Based on the received report, the sink generates appropriate routes (the flow rules) and send them to network.

Figure 5 compares our proposed method MSU-TSCH with a recently published scheduling method FTS-SDN [14].

As seen in Figure 5 (b), there is only one shared timeslot (timeslots one) when using the MSU-TSCH method as opposed to four timeslots (which are 1, 4, 7, 10) in FTS-SDN schedule. Allocated cells are reduced by three shared timeslots compared to TSCH schedule. Therefore, a drop in the occupancy rate at the slotframe level of up to 23%. MSU-TSCH allows to have an optimal number of shared time slots.

After the first phase of network discovery, the behavior of mobile nodes stabilizes. Where the sink is aware of every parent node that the mobile node uses. Each mobile node has a routing path and an assigned TSCH resources for each target. At this time the MSU-TSCH schedule has no influence in the end-to-end transmission time.

In MSU-TSCH, multicast cells have been included, the mobile node transmits three data packets in the same cell (timeslots 2) and receives three data packets in the same cell (timeslots 9) as shown in Figur 5 (b).

We take as an example the road (M->C->B->S). The TSCH Schedule slotframe in Figure 5 (a) and the MSU-TSCH slotframe in Figure 6 (b) both display all of the transmitting cells in red color. There are a total of 11 transmission states per slotframe in the FTS-SDN Schedule scenarios and 10 transmission states per slotframe in the MSU-TSCH.

Figure 6 shows the number of states for reception. On reception each node must listen to the traffic. At each shared timeslot, all nodes must listen, hence each shared timeslot is equivalent to four simple timeslots.

As show in Figure 6 (a), FTS-SDN presents four cell shared timeslots (1, 4, 7, 10) and 13 simple timeslots. Therefore, in total FTS-SDN presents 4*4+13= 29 reception states by slotframe.

However, our proposed MSU presents only 14 reception states by slotframe as shown in figure 6 (b). MSU bring a significant reduction percentage of up to 50% for power consumption in receive mode. This reduction is due to the decrease in number of shared timeslots (1, 4, 7, 10) as this is a factor that result in increased additional energy consumption across all nodes. Among the contributions of our approach, is that the schedule is dynamically updated according to the mobile node moving.

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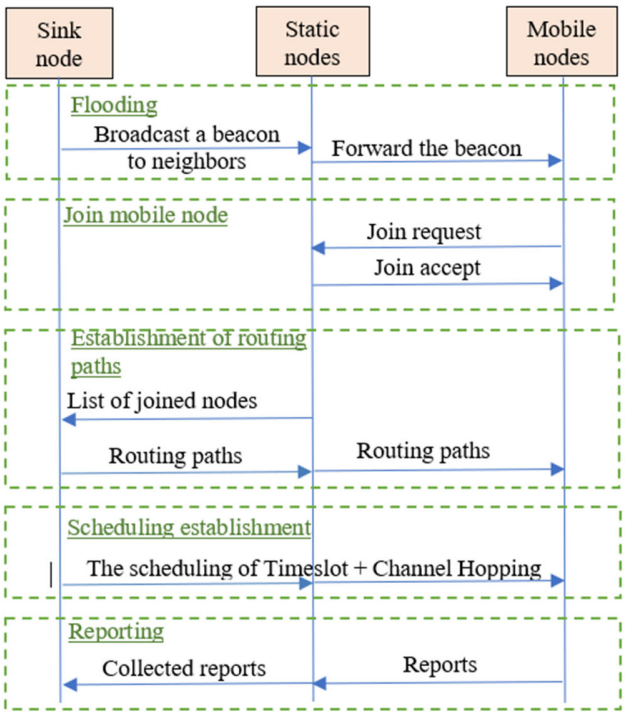

Fig.4 Messages exchange in MSU-TSCH

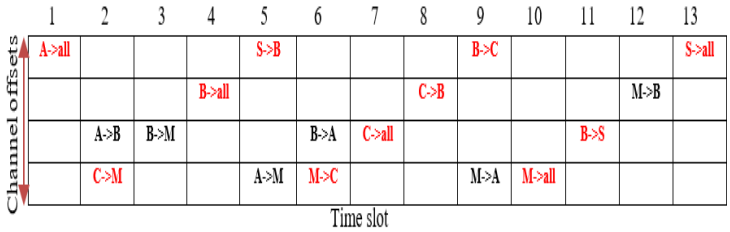

Fig. 5 (a): FTS-SDN Schedule transmission

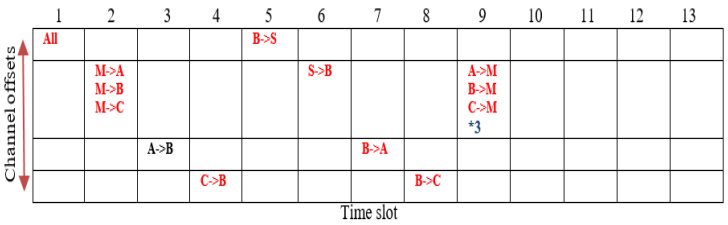

Fig. 5 (b): MSU-TSCH Schedule transmission

Fig. 5 Number of states for transmission

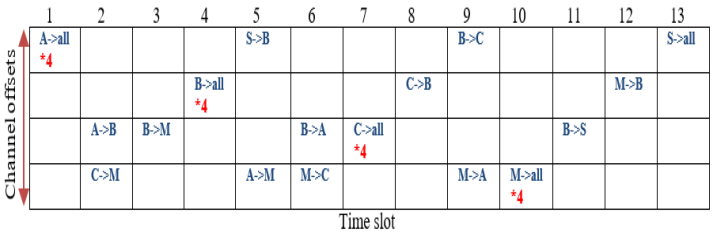

Fig. 6 (a): FTS-SDN Schedule reception

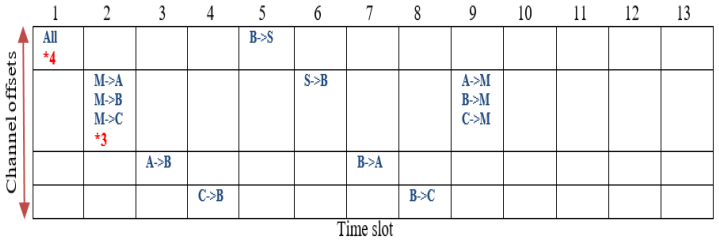

Fig. 6 (b): MSU-TSCH Schedule reception

Fig. 6 Number of states for reception

V. Performance Evaluation

This section displays the results of experiments performed with the COOJA (Contiki) simulator. Table 1 shows the parameters used during the simulations.

TABLE.1. SIMULATION PARAMETERS

Parameter	Value
Number of Node	8 fixed and 2 mobiles
Simulation time (s)	500 s
Number of channels	4
Radio medium	N=2,04; σ=6,7
Mobile node Speed	1 m/s
Timeslot (Ts)	10ms
Number of timeslots / superframe	9
Sending frequency	1 packet/s
Number of packets	500

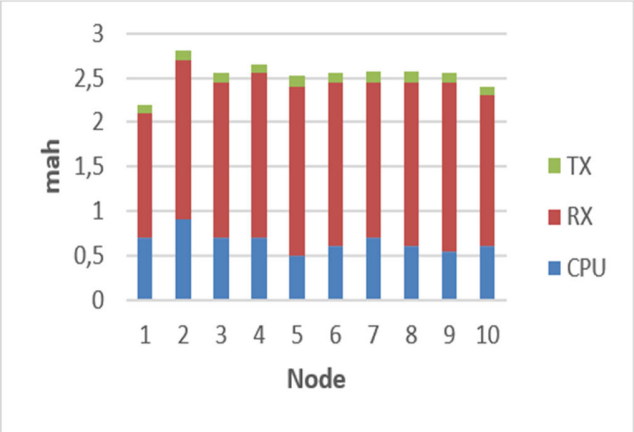

Fig. 7 (a): PRCOS

To calculate energy consumption, we count the number of transmissions and receiving packets at each node. The simulation of the topology lasted 500 s, giving researchers adequate time to look into long-term network behavior when collisions and buffer issues were present. Figure 7 shows the power consumption of each node depending on CPU, Radio Tx, and Radio Rx use for the same design.

The outcome is presented in Figure 7, which shows the power consumption at each node for the same architecture based on CPU, Radio Tx, and Radio Rx use. The consumption in this situation is roughly 2.5 mAh for the PRCOS algorithm and 1.5 for our proposed MSU-TSCH, and the majority of this energy is spent in reception, as shown in Figure 7.

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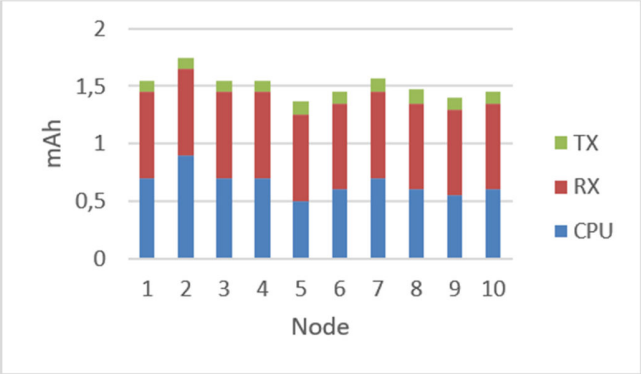

Fig. 7 (b): MSU-TSCH

MSU-TSCH employs four shared slots in Figure 7 (b), resulting in a total consumption of less than 1.65 mAh for all nodes and a significant decrease in the energy consumed for reception. This 40 percent reduction occurs when the PRCOS consumes 1.8 mAh and the MSU-TSCH consumes 0.75 mAh, as illustrated in Figure 7 (a) and (b), respectively. The CPU and Radio Tx values have no significant variances.

Due to the fact that each shared slot has just one transmitter node, the functioning of shared slots in PRCOS is quite particular. This necessitates an additional step on each node to configure the associated shared slots. This approach ensures transmission, unlike MSU-TSCH, which allows many nodes to transmit in the same shared slot, thereby causing collisions.

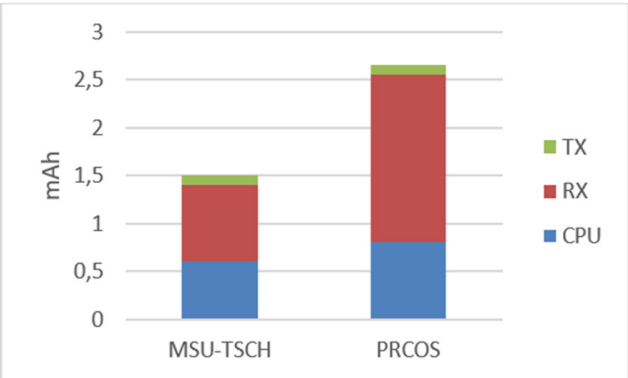

Fig. 7 (c) average Energy consumption

Fig.7: Energy consumption between PRCOS and MSU-TSCH

Figure 7 depicts the energy use (c). Because the number of shared slots is fewer, the power consumption at reception has been lowered compared to PRCOS. Because they must convey a lower quantity of control traffic, nodes will spend more time in receipt. Fewer transmissions minimize processing burden, resulting in lower CPU usage. This demonstrates that a lower quantity of control traffic may be used and that the loss of this sort of packet does not result in a performance loss.

Figure 8 depicts the number of control packets transmitted during a 500s simulation with beacon periods (Bp), Bp =1 s and report periods (Rp), Rp =2 s in both protocols. MSU-TSCH has lower PRCOS beacon levels and a larger proportion of reporting packet, that does not equal the amount of ACKs perfectly. This is caused by collisions in sharing slot, that impact all control messages.

As a result, MSU-TSCH has 12% more reports than ACKs. Changes in the parent of the mobile node cause missed reporting packet in PRCOS, equivalent about 4,2% of all report packets. Collisions in shared slots have no effect on MSU-TSCH performance since the MSU-TSCH does not require a quick topology update. Because control packets are used less frequently, the frequency of beacon and report packets may be decreased without sacrificing performance.

Figure 9 depicts the end-to-end latency when the mobile node's speed is 1 m/s and the schedules in Figure 3 are applied. The two methods have the predicted performance based on TSCH scheduling of 100% in this circumstance. The TSCH scheduling optimization utilized in MSU-TSCH, on the other hand, decreases the average end-to-end latency by 30%, from 55 ms in PRCOS to 30 ms. MSU-TSCH uses 2 times longer beacon and report durations than PRCOS, resulting in the same threshold being achieved with a lesser amount of control packets. It's challenging to adapt to topological changes induced by a mobile node because the node's speed and trajectory might fluctuate.

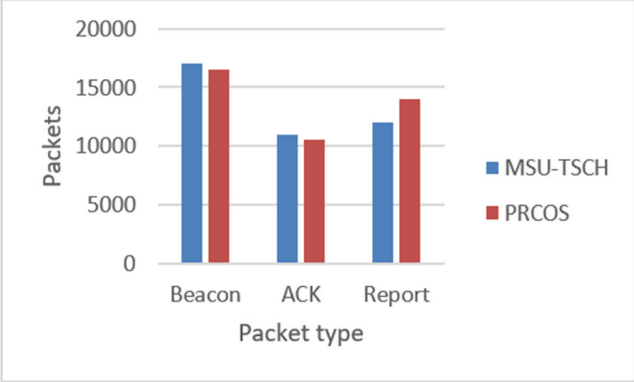

Fig. 8: During a 500 s simulation, the number of control packets transmitted was counted

As a consequence, the effect of speed on MSU-TSCH and PRCOS outcomes was studied. Because of the heavy reliance on traffic, a breakdown can occur at any moment, whereas continuous allocation of resources provides consistency by conserving the routes assigned and limiting the number of configurations offered by the Sink.

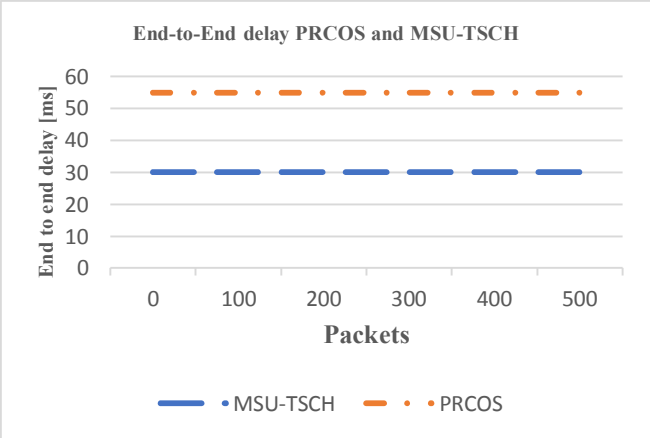

Fig.9. End to end delay between MSU-TSCH PRCOS

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Figure 10 depicts the end to end latency for the mobile node at various travel rates. In 0 the mobile travels at a rate of 1 m/s, which PRCOS and MSU-TSCH adapt to without losing any data. The PRCOS protocol works well in the initial transmission of packets so when rate is raised, but after a few packets, it begins to lose performance. This is because, unlike MSU-TSCH, which experiences losses during the first parent change but then stabilizes, the routes begin to become out of date as a result of the accumulation of queued packets.

With just a fast mobility speed of 4 m/s, the sink is unable to update the routes quickly enough, resulting in frequent losses in PRCOS Figure 10 (b) (around 15%). Figure 10 (d) shows that with MSU-TSCH, the route configuration only impacts the initial packets. After then, the behavior is pretty stable because no new adjustments are necessary, resulting in improved performance in the permanent mode.

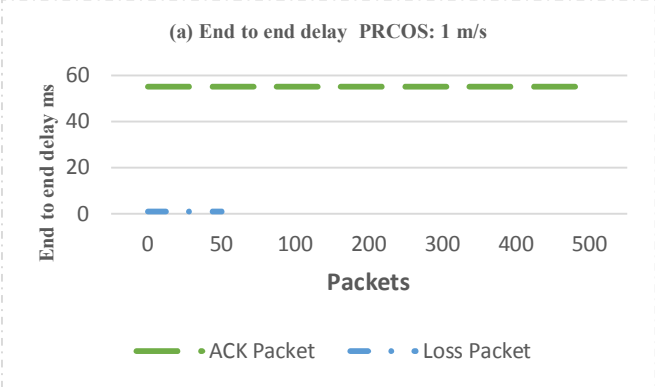

Fig. 10 (a): PRCOS: 1m/s

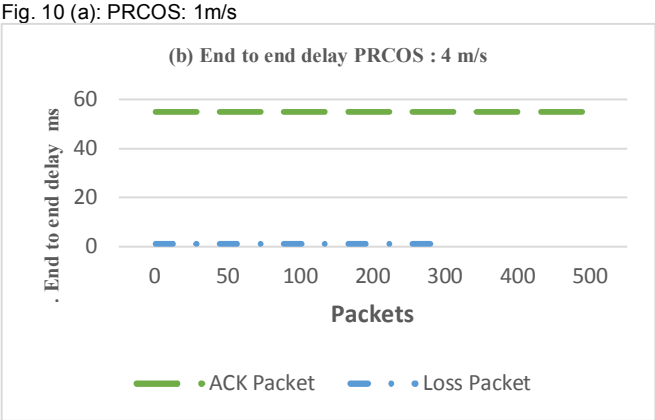

Fig. 10 (b): PRCOS: 4m/s

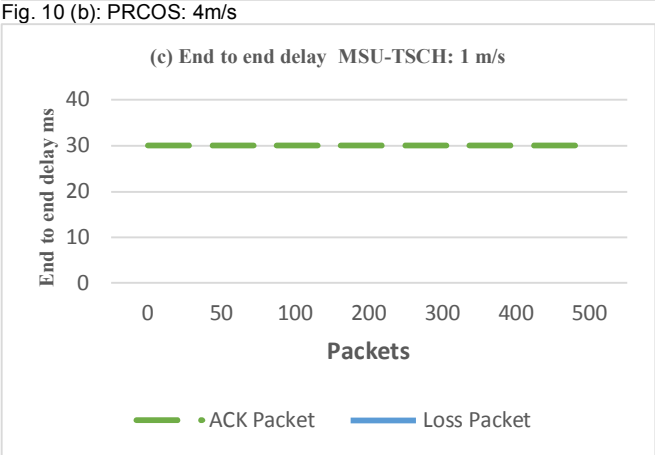

Fig. 10 (c): MSU-TSCH: 1m/s

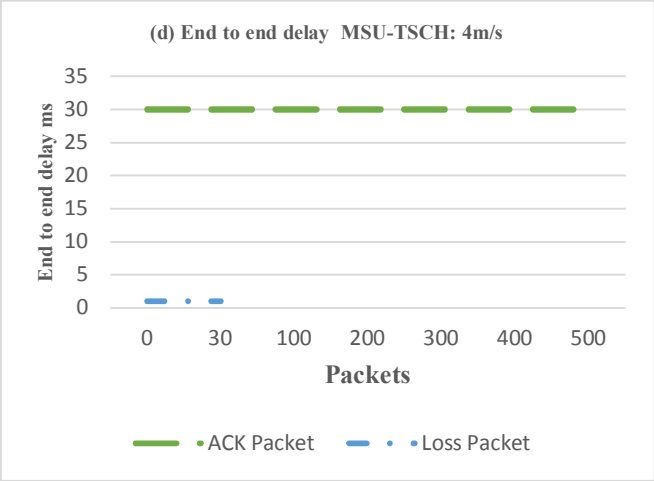

Fig. 10 (d):) MSU-TSCH: 4m/s

Fig.10. Comparison of MSU-TSCH vs PRCOS at various mobile node speeds

IV. Conclusion

Due to the dynamic of industrial context, WSNs must be able to swiftly react to physical and logical changes while preserving determinism and service quality. As a result, MSU-TSCH is presented for incorporating nodes in motion efficiently. It addresses the problems created by the routing protocols such as lack of convergence when the parent node moves, while preserving stringent QoS criteria. Because MSU-TSCH has shown to be independent from control traffic, this kind of traffic can be decreased by 30% compared to PRCOS. The lower dependency on control traffic that MSU-TSCH has proven.to have, allows this type of traffic to be reduced by 30% with respect to PRCOS.

Allocated cells are reduced by three shared timeslots compared to FTS-SDN schedule. Therefore, a drop in the occupancy rate and length at the slotframe level of up to 23%. MSU-TSCH allows to have an optimal number of shared time slots. Also, a reduction of the slotframe size goes from 13 timeslots for FSN-TSCH to 9 timeslots for our proposed protocol. This, together with the efficient use of shared slots, cuts energy usage at reception by up to 40%. The use of broadcast within mobile host reduces the issue delay of switching parents. As a consequence, the end-to-end latency has been decreased by 30%. Furthermore, because all parental nodes for each mobile node share the same transmit and receiving cell, slotframe utilization and duration are reduced by more than 30%.

References

[1] A. Saifullah, D. Gunatilaka, P. Tiwari, M. Sha, C. Lu, B. Li, C. Wu, and Y. Chen, "Schedulability Analysis under Graph Routing in WirelessHART Networks," in 2015 IEEE Real-Time Systems Symposium, San Antonio, TX, USA, 1-4 Dec. 2015, pp. 165–174.

[2] ISA-100.11a-2011:, "Wireless Systems for Industrial Automation: Process Control and Related Applications," International Society of Automation (ISA) Std., vol. 1, May 2011.

[3] H. Kurunathan, R. Severino, A. Koubaa, and E. Tovar, "Worst-case bound analysis for the time-critical MAC behaviors of IEEE 802.15.4e," in IEEE International Workshop on Factory Communication Systems (WFCS), Trondheim, Norway, 31 May - 2 June 2017.

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[4] W. Jerbi, O. Cheikhrouhou, H. Hamam, H. Trabelsi and A. Guermazi, "Incorporate mobility management into industrial wireless sensor networks", International Wireless Communications and Mobile Computing (IWCMC), 2022, Dubrovnik, Croatia, May 30 - June 3, 2022, pp. 475-480.
[5] T. Matsui and H. Nishi, "Time Slotted Channel Hopping Scheduling Based on the Energy Consumption of Wireless Sensor Networks", International Workshop on Advanced Motion Control (AMC), 2018, Tokyo, Japan, 10.1109/AMC.2019.8371162.
[6] A. K. Demir and S. Bilgili, "DIGEST: A decentralized divergecast scheduling algorithm for IEEE 802.15.4e TSCH in the Internet of Things", Wireless Netw., Oct. 2021, vol. 27, no. 7, pp. 4535–4550, ,doi:10.1007/s11276-021-02722-y.
[7] F. Righetti, C. Vallati, S.K. Das and G. Anastasi, "An Experimental Evaluation of the 6top Protocol for Industrial IoT Applications". In Proceedings of the 2019 IEEE Symposium on Computers and Communications (ISCC), Barcelona, Spain, 29 June–3 July 2019; pp. 1–6
[8] S. Kharb, A. Singhrova, "A survey on network formation and scheduling algorithms for time slotted channel hopping in industrial networks", Journal of Network and Computer Applications, 2019, pp 59-87,
[9] J. Jung, D. Kim, J. Hong, J. Kang, Y. Yi, "Parameterized slot scheduling for adaptive and autonomous TSCH networks". In Proceedings of the IEEE INFOCOM 2018, Honolulu, HI, USA, pp.76-81, DOI: 10.1109/INFCOMW.2018.8407031.
[10] S. Rekik, N. Baccour, M. Jmaiel, K. Drira, L.A. Grieco, "Autonomous and traffic-aware scheduling for TSCH networks". Comput. Netw. 2018, vol. 135, pp. 201–212, doi.org/10.1016/j.comnet.2018.02.023.
[11] M. Nsabagwa, J. Muhumuza, R. Kasumba, J.S. Otim, R. Akol, "Minimal Idle-Listen Centralized Scheduling in TSCH Wireless Sensor Networks", 41st International Conference on Telecommunications and Signal Processing (TSP), Athens, Greece, 4–6 July 2018; pp. 1–5.
[12] M.R. Palattella, N. Accettura, M. Dohler, L.A. Grieco, G. Boggia, "Traffic Aware Scheduling Algorithm for reliable low-power multi-hop IEEE 802.15.4e networks", 2012, IEEE 23rd International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), Sydney, Australia, September 2012, pp. 327–332.
[13] Y. Jin, P. Kulkarni, J. Wilcox, M. Sooriyabandara, "A centralized scheduling algorithm for IEEE 802.15.4e TSCH based industrial low power wireless networks", In Proceedings of the 2016 IEEE Wireless Communications and Networking Conference, Doha, Qatar, 3–6 April 2016; pp. 1–6.
[14] L. Lo Bello, A. Lombardo, S. Milardo, G. Patti, M. Reno, "Experimental Assessments and Analysis of an SDN Framework to Integrate Mobility Management in Industrial Wireless Sensor Networks", 2020, IEEE Transactions on Industrial Informatics, DOI 10.1109/TII.2020.2963846.
[15] T. Devaja, D. Bajovic, D. Vukobratovic, G. Gardaševic, "Scheduling in 6TiSCH Networks via Max-Product Message-Passing". In Proceedings of the IEEE EUROCON 2019—18th International Conference on Smart Technologies, Novi Sad, Serbia, 1–4 July 2019; pp. 1–6.
[16] H. Park, H. Kim, K.T. Kim, S. Kim, P. Mah, "Frame-Type-Aware Static Time Slotted Channel Hopping Scheduling Scheme for Large-Scale Smart Metering Networks". IEEE Access 2019, vol. 7, pp. 2200–2209.
[17] O. Gaddour, A. Koubäa, R. Rangarajan, O. Cheikhrouhou, E. Tovar, and M. Abid, "Co-RPL: RPL routing for mobile low power wireless sensor networks using Corona mechanism". In Proceedings of the 9th IEEE international symposium on industrial embedded systems, Pisa, Italy, June 18-20, 2014, pp. 200-209.
[18] F. O. Santos, V. S. PAYÁ, J. S. Blanes and T. A. Albero, "Multicast Scheduling in SDN WISE to Support Mobile Nodes in Industrial Wireless Sensor Networks", 2021 IEEE ACCESS, Vol 9, pp 141651-141666.

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